



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: GRID-INDIA/NLDC/SO/Daily PSP Report

दिनांक: 17.12.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 16.12.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-38(1) के प्रावधान के अनुसार, दिनांक 16-दिसंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 38(1) of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 16.12.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 17-Dec-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 19:00 hrs; from RLDCs)	58383	63063	41914	20952	2659	186971
Peak Shortage (MW)	0	0	0	0	0	0
Energy Met (MU)	1221	1466	1035	428	48	4198
Hydro Gen (MU)	118	39	81	18	14	269
Wind Gen (MU)	10	89	49	-	-	149
Solar Gen (MU)*	153.29	98.68	121.59	3.61	1.20	378
Energy Shortage (MU)	1.66	0.00	0.00	0.00	0.00	1.66
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	66219	73581	51723	21953	2791	212807
Time Of Maximum Demand Met	10:35	09:56	09:39	18:05	17:24	09:42

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.090	1.79	3.38	13.70	18.88	65.04	16.09

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	8991	0	159.6	65.5	-0.7	536	0.00
	Haryana	8903	0	160.3	103.8	-0.9	799	0.00
	Rajasthan	17826	0	331.0	126.6	1.9	677	0.00
	Delhi	4394	0	76.8	76.0	-1.4	346	0.00
	UP	19642	0	338.8	109.3	-2.4	256	0.00
	Uttarakhand	2378	0	43.8	32.2	0.8	161	0.04
	HP	2155	0	38.6	31.7	-2.9	220	0.00
	J&K(UT) & Ladakh(UT)	3032	0	63.7	59.0	-0.2	515	1.62
	Chandigarh	258	0	4.3	4.3	0.0	22	0.00
	Railways_NR IST'S	203	0	4.1	3.7	0.4	36	0.00
WR	Chhattisgarh	4743	0	97.1	37.7	-0.8	288	0.00
	Gujarat	21107	0	402.8	201.7	-0.6	1044	0.00
	MP	18555	0	336.8	208.0	-1.7	892	0.00
	Maharashtra	26917	0	554.7	190.4	-1.2	565	0.00
	Goa	693	0	13.9	11.2	2.2	41	0.00
	DNHDDPDCL	1283	0	29.1	29.4	-0.3	42	0.00
	AMNSIL	783	0	17.5	7.0	0.5	269	0.00
	BALCO	529	0	12.6	12.7	-0.1	527	0.00
	RIL JAMNAGAR	85	0	1.5	1.8	-0.3	0	0.00
SR	Andhra Pradesh	9978	0	191.8	56.5	0.7	787	0.00
	Telangana	12901	0	242.9	153.6	-0.8	757	0.00
	Karnataka	13597	0	237.7	55.0	-2.6	467	0.00
	Kerala	4192	0	78.9	63.4	-0.1	440	0.00
	Tamil Nadu	14695	0	275.8	159.4	-1.9	766	0.00
	Puducherry	407	0	8.3	8.1	-0.2	58	0.00
ER	Bihar	5061	0	88.5	78.0	-1.2	236	0.00
	DVC	3183	0	69.3	-39.9	1.3	335	0.00
	Jharkhand	2012	0	38.3	26.0	0.3	255	0.00
	Odisha	4836	0	91.9	28.5	-2.1	488	0.00
	West Bengal	7100	0	137.6	9.4	-1.5	262	0.00
	Sikkim	117	0	1.8	1.6	0.2	43	0.00
	Railways_ER IST'S	23	0	0.2	0.3	-0.1	3	0.00
NER	Arunachal Pradesh	167	0	3.0	2.8	0.0	46	0.00
	Assam	1527	0	26.5	18.8	0.6	129	0.00
	Manipur	228	0	2.9	3.7	-0.8	36	0.00
	Meghalaya	366	0	6.4	5.8	-0.4	120	0.00
	Mizoram	143	0	2.6	1.7	-0.1	9	0.00
	Nagaland	163	0	2.5	2.4	-0.1	28	0.00
	Tripura	231	0	4.4	3.2	0.3	38	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	-6.2	-3.7	-13.8	-13.1
Day Peak (MW)	-432.4	-224.0	-812.0	-769.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	233.8	-220.0	160.7	-178.6	4.2	0.0
Actual(MU)	230.1	-228.6	160.4	-172.8	3.0	-7.8
O/D/U/D(MU)	-3.6	-8.6	-0.2	5.8	-1.1	-7.8

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	4874	8414	6948	4998	518	25752	45
State Sector	7798	12961	7374	2629	191	30952	55
Total	12672	21375	14322	7627	709	56704	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	706	1480	568	654	10	3418	75
Lignite	26	7	33	0	0	66	1
Hydro	118	39	81	18	14	269	6
Nuclear	32	68	67	0	0	166	4
Gas, Naptha & Diesel	2	13	2	0	25	42	1
RES (Wind, Solar, Biomass & Others)	190	190	198	7	1	586	13
Total	1074	1797	948	679	50	4548	100

Share of RES in total generation (%)	17.69	10.60	20.85	1.02	2.42	12.89
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	31.61	16.53	36.42	3.65	30.00	22.46

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.016
Based on State Max Demands	1.049

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	212807	9:42	404
Non-Solar hr	196583	18:30	179

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 17-Dec-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	249	0.0	6.3	-6.3
3	765 kV	GAYA-VARANASI	2	0	1000	0.0	15.2	-15.2
4	765 kV	SASARAM-FATEHPUR	1	0	531	0.0	8.6	-8.6
5	765 kV	GAYA-BALIA	1	0	739	0.0	11.6	-11.6
6	400 kV	PUSAULI-VARANASI	1	0	224	0.0	3.9	-3.9
7	400 kV	PUSAULI -ALLAHABAD	1	0	129	0.0	2.2	-2.2
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	602	0.0	8.2	-8.2
9	400 kV	PATNA-BALIA	2	0	893	0.0	16.2	-16.2
10	400 kV	NAUBATPUR-BALIA	2	0	299	0.0	5.1	-5.1
11	400 kV	BIHARSHARIFF-BALIA	2	0	287	0.0	4.5	-4.5
12	400 kV	MOTIHARI-GORAKHPUR	2	0	540	0.0	9.2	-9.2
13	400 kV	BIHARSHARIFF-VARANASI	2	0	0	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	66	0.0	0.7	-0.7
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.1	0.0	0.1
16	132 kV	GARWAH-RIHAND	1	30	0	0.5	0.0	0.5
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.6	91.4	-90.8
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	55	1342	0.0	15.8	-15.8
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	284	1275	0.0	6.6	-6.6
3	765 kV	JHARSUGUDA-DURG	2	0	810	0.0	13.9	-13.9
4	400 kV	JHARSUGUDA-RAIGARH	4	0	746	0.0	10.4	-10.4
5	400 kV	RANCHI-SIPAT	2	0	1	0.0	0.0	0.0
6	220 kV	BUDHIPADAR-RAIGARH	1	23	112	0.0	1.2	-1.2
7	220 kV	BUDHIPADAR-KORBA	2	118	76	0.6	0.0	0.6
ER-WR						0.6	47.9	-47.3
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	528	0.0	9.0	-9.0
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1646	0.0	29.9	-29.9
3	765 kV	ANGUL-SRIKAKULAM	2	0	2884	0.0	53.5	-53.5
4	400 kV	TALCHER-I/C	2	1738	682	12.7	0.0	12.7
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	92.4	-92.4
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	248	0	3.8	0.0	3.8
2	400 kV	ALIPURDUAR-BONGAIGAON	2	447	0	7.6	0.0	7.6
3	220 kV	ALIPURDUAR-SALAKATI	2	93	0	1.5	0.0	1.5
ER-NER						12.8	0.0	12.8
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	666	0	16.4	0.0	16.4
NER-NR						16.4	0.0	16.4
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	3545	0.0	53.5	-53.5
2	HVDC	VINDHYACHAL B/B	-	0	55	0.0	1.2	-1.2
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1914	0.0	26.6	-26.6
4	765 kV	GWALIOR-AGRA	2	0	2011	0.0	25.6	-25.6
5	765 kV	GWALIOR-PHAGI	2	494	1840	1.0	23.3	-22.3
6	765 kV	JABALPUR-ORAI	2	0	980	0.0	27.3	-27.3
7	765 kV	GWALIOR-ORAI	1	908	0	14.0	0.0	14.0
8	765 kV	SATNA-ORAI	1	0	1053	0.0	19.8	-19.8
9	765 kV	BANASKANTHA-CHITORGARH	2	1285	476	8.6	1.1	7.5
10	765 kV	VINDHYACHAL-VARANASI	2	0	2123	0.0	33.3	-33.3
11	400 kV	ZERDA-KANKROLI	1	349	55	2.3	0.1	2.2
12	400 kV	ZERDA -BHINMAL	1	241	38	1.6	0.1	1.6
13	400 kV	VINDHYACHAL -RIHAND	1	971	0	21.8	0.0	21.8
14	400 kV	RAPP-SHUJALPUR	2	559	319	3.6	1.6	2.0
15	400 kV	NEEMUCH-Chittorgarh	2	583	321	3.7	1.4	2.3
16	220 kV	BHANPURA-RANPUR	1	0	203	0.0	3.3	-3.3
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	1.9	-1.9
18	220 kV	MEHGAON-AURAIYA	1	118	0	1.3	0.0	1.3
19	220 kV	MALANPUR-AURAIYA	1	80	7	0.8	0.0	0.8
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						58.6	220.0	-161.4
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1007	0.0	21.1	-21.1
2	HVDC	RAIGARH-PUGALUR	2	0	3505	0.0	51.2	-51.2
3	765 kV	SOLAPUR-RAICHUR	2	1755	1110	7.8	8.1	-0.4
4	765 kV	WARDHA-NIZAMABAD	2	0	2246	0.0	31.2	-31.2
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2353	0.0	35.5	-35.5
6	400 kV	KOLHAPUR-KUDGI	2	1662	0	22.7	0.0	22.7
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	0	0.0	0.0	0.0
9	220 kV	XELDEM-AMBEWADI	1	0	139	2.4	0.0	2.4
WR-SR						32.9	147.0	-114.1

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)		
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	140	-126	-9	-0.22	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	183	-151	-31	-0.75	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-236	-134	-199	-4.78	
	NER	132kV GELEPHU-SALAKATI	-24	-14	-20	-0.48	
	NER	132kV MOTANGA-RANGIA	7	0	2	0.04	
NEPAL	NR	NEPAL IMPORT (FROM UP)	-64	0	0	0.00	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-64	0	-39	-0.94	
	ER	NEPAL IMPORT (FROM BIHAR)	0	0	0	0.00	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-224	2	-115	-2.76	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-756	-490	-529	-12.68	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-769	-394	-544	-13.05	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-56	0	-46	-1.10	

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	7.28	0.00	0.00	0.06	0.00	0.00	7.34
Nepal	0.13	0.00	0.67	0.00	0.00	2.27	0.00	0.00	3.07
Bangladesh	13.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.21
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	13.34	0.00	7.95	0.00	0.00	2.33	0.00	0.00	23.62

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.63
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.63

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.63	0.00	-7.28	0.00	0.00	-0.06	0.00	0.00	-6.71
Nepal	-0.13	0.00	-0.67	0.00	0.00	-2.27	0.00	0.00	-3.07
Bangladesh	-13.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-13.21
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-12.71	0.00	-7.95	0.00	0.00	-2.33	0.00	0.00	-22.99

15 Min (INSTANTANEOUS) ALL INDIA GRID FREQUENCY, GENERATION & DEMAND MET (SCADA DATA)												
TIME	FREQUENCY (Hz)	DEMAND MET (MW)	NUCLEAR (MW)	WIND (MW)	SOLAR (MW)	HYDRO** (MW)	GAS (MW)	THERMAL (MW)	OTHERS* (MW)	NET DEMAND MET (MW)	TOTAL GENERATION (MW)	NET TRANSNATIONAL EXCHANGE (MW)
		(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I=A-(C+D))	(J=B+C+D+E+F+G+H)	(K)
0:00	50.16	137645	6563	8846	4759	7532	1564	113554	2969	124040	145787	-1025
0:15	50.06	137325	6565	8830	4758	6978	1561	114501	2966	123737	146159	-1063
0:30	50.11	136249	6569	8724	4759	6859	1557	112869	2955	122766	144292	-1055
0:45	50.01	135645	6562	8684	4773	6746	1556	112435	2990	122188	143746	-1068
1:00	50.01	134711	6553	8559	4774	6514	1563	112085	2983	121378	143031	-1078
1:15	50.01	134001	6562	8460	4773	6424	1564	112526	3008	120768	143317	-1092
1:30	50.01	133293	6566	8414	4773	6399	1562	111087	2948	120106	141749	-1021
1:45	50.01	132699	6562	8322	4757	6186	1557	110648	3016	119620	141048	-1000
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